

UDINE RIVOLTO, Italy

WMO: 160450

Lat: 45.979N

Lon: 13.049E

Elev: 179

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 90-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 23.4	(c) 26.3	(d) 5.2	(e) 7.2	(f) 37.8	(g) 10.0	(h) 9.2	(i) 34.4	(j) 22.8	(k) 40.3	(l) 19.9	(m) 40.1	(n) 5.0	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	91.0	73.0	88.2	72.1	86.1	71.3	76.3	85.7	74.5	83.8	73.0	82.2	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	125.0	82.1	71.5	117.2	79.8	70.0	111.1	78.3	39.8	86.0	38.1	83.9	36.7	82.3	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.4	(b) 15.0	(c) 12.2	(e) 17.4	(f) 95.2	(g) 3.3	(h) 2.0	(i) 15.0	(j) 96.7	(k) 13.1	(l) 97.9	(m) 11.2	(n) 99.0	(o) 8.8	(p) 100.5		
(a) 18.4	(b) 15.0	(c) 12.2	(e) 16.2	(f) 79.3	(g) 2.7	(h) 2.4	(i) 14.3	(j) 81.1	(k) 12.7	(l) 82.5	(m) 11.2	(n) 83.8	(o) 9.2	(p) 85.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.0	40.2	41.8	49.2	55.2	64.9	70.3	74.7	74.7	65.2	57.5	48.5	40.7
	DBStd	13.62	5.06	6.02	5.82	5.37	5.76	5.70	5.05	4.74	5.01	5.94	6.76	5.41
	HDD50	1048	305	235	88	12	0	0	0	0	0	10	106	292
	HDD65	3839	767	651	490	297	69	16	2	2	58	239	494	754
	CDD50	3607	2	4	63	168	463	609	766	767	456	243	63	3
	CDD65	921	0	0	0	3	67	174	302	304	64	7	0	0
	CDH74	8151	0	0	2	49	482	1409	2893	2853	405	58	0	0
	CDH80	2723	0	0	0	2	85	413	1087	1083	51	2	0	0
Wind	WSAvg	4.8	4.7	4.5	5.7	5.4	5.1	4.6	4.7	4.4	4.4	4.7	4.4	4.9
Precipitation	PrecAvg	42.6	2.5	2.2	2.5	4.1	3.4	3.3	3.3	3.5	4.7	4.7	4.6	3.1
	PrecMax	59.8	9.0	9.5	7.9	10.0	7.9	8.9	6.2	8.3	12.5	11.9	13.9	7.1
	PrecMin	28.9	0.2	0.1	0.0	0.6	0.5	0.6	1.1	1.0	1.3	0.4	0.3	0.2
	PrecStd	8.1	2.2	2.8	2.1	2.4	1.9	1.8	1.5	2.3	2.8	3.2	2.9	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	65.1	73.0	80.3	87.4	92.5	94.1	94.8	84.1	79.4	68.5	60.0
		MCWB	50.2	51.0	54.7	62.1	70.4	75.5	73.7	73.1	69.4	64.8	59.2	54.1
	2%	DB	53.8	59.1	68.0	76.5	82.1	87.6	90.5	91.4	81.3	75.2	64.5	55.2
		MCWB	47.9	48.0	53.7	60.8	66.8	72.2	73.0	72.9	67.6	62.7	58.2	48.9
	5%	DB	51.6	55.6	63.7	71.8	78.8	84.7	88.4	88.2	78.6	71.7	61.1	51.5
		MCWB	46.7	46.8	51.9	58.1	64.6	70.9	72.2	72.0	66.4	61.1	56.4	46.1
	10%	DB	49.2	52.1	59.3	67.7	76.5	81.7	86.1	85.9	75.9	67.9	58.8	49.1
		MCWB	44.7	45.5	50.6	56.3	63.7	69.3	71.7	71.0	65.4	60.5	54.1	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.1	57.4	63.6	72.0	76.5	78.6	78.9	73.6	67.9	62.6	56.7
		MCDB	56.1	60.1	67.9	77.0	82.0	86.3	89.2	87.0	79.3	72.9	67.4	59.0
	2%	WB	50.2	50.2	55.4	61.3	69.9	74.4	76.4	76.3	71.0	65.6	59.2	49.8
		MCDB	52.4	57.3	63.4	73.8	79.0	83.6	86.2	85.6	77.9	71.5	62.5	51.9
	5%	WB	47.9	48.3	53.8	59.4	67.7	72.6	74.5	74.4	69.0	63.6	56.9	47.6
		MCDB	50.6	53.7	60.4	69.8	75.5	81.5	84.1	83.7	75.2	69.4	60.5	50.3
	10%	WB	45.3	46.3	52.2	57.4	65.7	71.1	72.9	72.8	67.0	61.2	54.8	45.8
		MCDB	48.1	50.5	57.8	65.2	73.0	79.5	82.4	82.1	73.1	66.5	58.2	48.6
Mean Daily Temperature Range	5% DB	MDBR	14.4	17.4	18.1	17.8	18.0	17.3	19.2	19.8	18.5	17.2	15.7	15.0
		MCDBR	18.3	24.0	24.5	24.9	21.5	20.7	21.3	22.1	20.8	23.7	18.3	18.8
		MCWBR	13.1	15.2	13.9	12.6	9.7	9.0	8.7	9.7	10.0	13.2	12.3	13.8
		MCDDBR	13.6	19.6	20.0	21.8	17.0	18.7	19.0	20.5	17.2	17.6	13.6	14.0
	5% WB	MCWBR	11.1	13.6	12.6	12.4	9.7	9.7	9.4	10.5	10.7	11.7	10.0	11.4
Clear-Sky Solar Irradiance	taub	0.335	0.362	0.410	0.453	0.462	0.478	0.465	0.456	0.435	0.415	0.368	0.331	
	taud	2.406	2.322	2.195	2.102	2.127	2.132	2.177	2.215	2.242	2.259	2.339	2.432	
	Ebn at Noon	240	254	256	255	257	253	255	252	246	230	222	228	
	Edh at Noon	23	30	40	47	48	48	45	42	37	32	25	21	
All-Sky Solar Radiation	RadAvg	441	714	1101	1425	1723	1947	1980	1707	1252	795	457	382	
	RadStd	62	113	141	170	185	124	117	147	119	90	73	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (59 neighbors)	+0.81	N/A	N/A	+1.22	+0.72	N/A	N/A	-221	+169	+101

Nomenclature: See separate page